SOLUTIONS
SECTION-I

Single Correct Answer Type **12 Q. [3 M (–1)]**

1. Ans. (B)

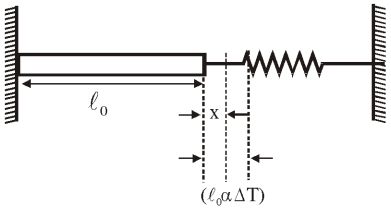
Sol. $\tan\theta = \frac{AP}{CP}$; $\ell n(\tan\theta) = \ell n(AP) - \ell n(CP)$

$\frac{\sec^2\theta}{\tan\theta}\theta d\theta = \frac{d(AP)}{AP} - \frac{d(CP)}{CP} = (\alpha_y - \alpha_x)\Delta T$. So dθ is negative.

2. Ans. (A)

Sol. Change in length of rod = $(\ell_0\alpha\Delta T - x)$

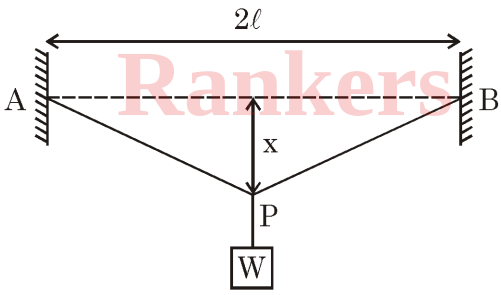
$kx = \left(\frac{\ell_0\alpha\Delta T - x}{\ell_0}\right) yA \Rightarrow x[k\ell_o + YA] = \ell_o\alpha\Delta TYA$



$\therefore \frac{F}{A} = \frac{kx}{A} = \frac{k\ell_o\alpha\Delta TY}{k\ell_o + YA}$

3. Ans. (C)

Sol.



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$AP = BP = (x^2 + \ell^2)^{1/2}$

$\Delta\ell = 2(x^2 + \ell^2)^{1/2} - 2\ell$

$= 2\ell\left[\left(1 + \frac{x^2}{\ell^2}\right)^{1/2} - 1\right] = 2\ell\left[1 + \frac{1}{2}\frac{x^2}{\ell^2} - 1\right]$ $(\because (1+x)^n = 1 + nx \text{ if } x << 1)$

$= \frac{x^2}{\ell}$ $\because \text{Strain} = \frac{\text{Change in length}}{\text{Original length}}$

$= \frac{(x^2/\ell)}{2\ell} = \frac{x^2}{2\ell^2}$

4. Ans. (B)

Sol. Suppose m kg steam is required per hour

Heat is released by steam in following three steps

(i) When 150°C steam $\xrightarrow{Q_1}$ 100°C steam $Q_1 = mc_{steam}\Delta\theta = m \times 1\ (150–100) = 50\ m$ cal

(ii) When 150°C steam $\xrightarrow{Q_2}$ 100°C water $Q_2 = mL_v = m\times 540 = 540\ m$ cal

(iii) When 100 °C water $\xrightarrow{Q_3}$ 90°C water $Q_3 = mc_W\Delta\theta = m \times 1 \times (100 - 90) = 10\ m$ cal

Hence total heat given by the steam $Q = Q_1 + Q_2 + Q_3 = 600$ mcal ....(i)

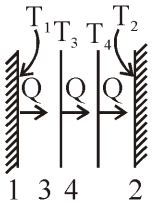
Heat taken by 10 kg water $Q' = mc_W\Delta\theta = 10 \times 10^3 \times 1 \times (80–20) = 600 \times 10^3$ cal

Hence $Q = Q' \Rightarrow 600\ m = 600 \times 10^3 \Rightarrow m = 10^3$ gm = 1 kg

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5. Ans. (B)

Sol.



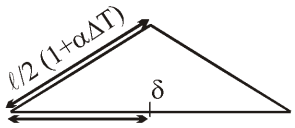
$$Q = \sigma\left(T_1^4 - T_3^4\right) W/m^2 = \sigma\left(T_3^4 - T_4^4\right) W/m^2 = \sigma\left(T_4^4 - T_2^4\right) W/m^2$$

$$3Q = \sigma\left(T_1^4 - T_2^4\right) = Q_0 W/m^2$$

$$\eta = \frac{Q}{Q_0} = \frac{1}{3}$$

6. Ans. (A)

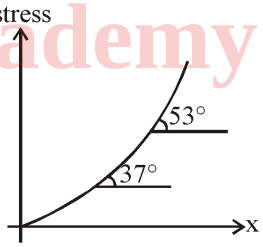
Sol. $\delta^2 = (\ell/2(1+\alpha\Delta T))^2 - (\ell/2)^2$



7. Ans. (A)

Sol. $\tan\theta = \frac{d(\text{stress})}{dx}$ (slope)

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$$= \frac{dF/A}{dx} = \frac{(dm)a/A}{dx} = \frac{(\rho \cdot d(\text{vol.}))a}{A\ dx}$$

$$\tan\theta = \rho a$$

$$\frac{\tan 37^\circ}{\tan 53^\circ} = \frac{\rho_1}{\rho_2} = \frac{9}{\rho_2} \qquad \Rightarrow \rho_2 = 16\ \text{gm/cm}^3$$

8. Ans. (B)

Sol. $\frac{dT}{dt} = \frac{\sigma e A}{ms}\left(T^4 - T_S^4\right)$

$\frac{dT}{dt} \propto \frac{A}{m}$ $\quad (m = \rho V)$

for disc (pizzas) $\quad m \propto A \quad$ so $\frac{dT}{dt}$ is same for both

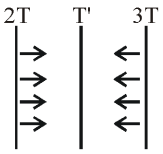
for gulab jamuns (sphere) → small (A, m); big (4A, 8m)

$\left(\frac{dT}{dt}\right)_{small} > \left(\frac{dT}{dt}\right)_{big}$ (for Gulab jamun)

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9. Ans. (C)

Sol.



Let temperature of middle plate = T'

$k(2T)^4 + k(3T)^4 = 2k(T')^4$

$$T' = \left(\frac{97}{2}\right)^{1/4} T$$

10. Ans. (A)

Sol. Since $t = \frac{\rho L}{2K\theta}(x_2^2 - x_1^2)$

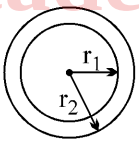
$$\therefore\ t = \frac{\rho L}{2K\theta}(x^2 - y^2) = \frac{\rho L(x+y)(x-y)}{2K\theta}$$

11. Ans. (D)

Sol. $\int dR = \int \frac{dr}{4\pi r^2 K} = \frac{1}{4\pi K}\left[\frac{r_2 - r_1}{r_2 r_1}\right]$ [R = thermal resistance]

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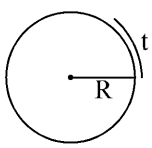
$$\dot{Q} = \frac{4\pi K\Delta\theta}{\left(\frac{r_2 - r_1}{r_1 r_2}\right)} = \frac{4\pi K\Delta\theta}{\left(\frac{t}{r^2}\right)}$$

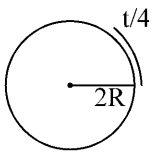


$$\frac{mL}{time} = \frac{4\pi K\Delta\theta}{\left(\frac{t}{r^2}\right)} \qquad \left[m = \rho \times \frac{4}{3}\pi r^3\right]$$

$$\frac{\rho L}{time} = \left(\frac{K}{t\,r}\right) \times \text{constant}$$

$$time \propto \frac{t\,r}{K}$$



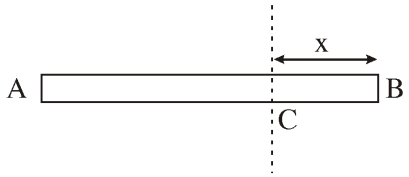


$$\frac{25}{16} = \frac{\frac{t}{4} 2r K_s}{t\,r K_L} = \frac{1 K_s}{2 K_L};\ \frac{K_L}{K_s} = \frac{8}{25}$$

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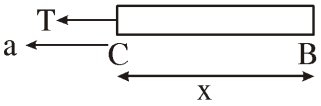
12. Ans. (B)

Sol.



at any distance x from edge let there be a cross section for which FBD is given:

for complete rod : $ma = F \Rightarrow a = \frac{F}{m}$ ...(1)



for portion BC : $m_{BC}\, a = T \Rightarrow T = \left(\frac{mx}{L}\right) a;\ T = \left(\frac{mx}{L}\right)\frac{F}{m} = \frac{F}{L}x \Rightarrow \text{stress} = \Rightarrow \frac{Fx}{AL}$

(C) $\frac{d\ell}{dx} = \frac{stress}{Y} = \frac{\frac{Fx}{LA}}{Y} = \frac{Fx}{LAY} \Rightarrow \Delta\ell = \int_0^{\ell} \frac{Fx}{LAY}dx \Rightarrow \frac{\Delta\ell}{L} = \frac{F}{2AY}$

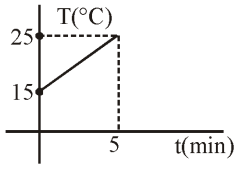
(D) Energy stored $= \int_0^{L} \frac{1}{2}\,\frac{F^2}{L^2A^2Y}\,x^2\,dx\,A = \frac{\frac{1}{2}F^2A}{L^2A^2Y}\left(\frac{L^3}{3}\right) = \frac{F^2L}{6AY}$

Multiple Correct Answer Type — 6 Q. [4 M (–1)]

13. Ans. (A, C)

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Sol. In final state, $\frac{dQ_e}{dt} = k(45-15) = 12$



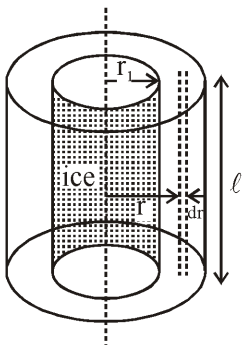
$K = \frac{2}{5}$; at 20°C, $\frac{dQ_e}{dt} = k(20-15) = \frac{2}{5}\times 5 = 2\text{watt}.$

$\text{m.s.}\ \frac{dT}{dt} = 12 - \frac{2}{5}(T-15) \Rightarrow \text{m.s.}\ \frac{dT}{dt} = 18 - \frac{2}{5}T$

$\text{m.s.}\ \int_{15}^{25}\frac{dT}{18-\frac{2}{5}T} = \int_0^{300} dt \Rightarrow s = \frac{240}{\ell n\left(\frac{3}{2}\right)}\ J/kg-°C$

14. Ans. (A,B)

Sol.



$R_{eq} = \int_{r_1}^{r_2}\frac{dr}{k.2\pi r\ell} = \frac{\ell n2}{k.2\pi\ell}$

$\therefore$ Heat current $= \frac{T_S - T_{ice}}{R_{eq}} = \frac{40}{\ell n2}k2\pi\ell = 80\pi\ J/sec$

Also, $\frac{dm_{ice}}{dt}L_f = 80\pi$

$\Rightarrow \frac{dm_{ice}}{dt} = \frac{80\pi}{80\times 4.2\times 1000} = \frac{\pi}{4200}\ kg/sec$

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15. Ans. (A,D)

16. Ans. (B, D)

17. Ans. (A,D)

Sol. Heat required to melt ice = $50 \times 1/2 \times (0 - (-10)) + 50 \times 80 = 4250$ cal.

heat av. from condension steam = $10 \times 540 = 5400$ cal

$\Rightarrow$ whole ice can melt. To rais temperature of system from 0° to 100°.

Heat required = $50 \times 1 \times (100-0) + 1.5 \times 1 \times (100-0)$

= 5150 cal. × not possible.

So, finally the substance is in from of water

$5400 + 10 \times 1 \times (100 - T) = 4250 + 50 \times 1\ (T - 0) + 1.5 \times 1(T - 0)$

$2150 = 61.5\ T$

$$T = \frac{2150}{61.5} = 35°C$$

18. Ans. (B,D)

Sol. $$\frac{P}{4\pi R_S^2} = \sigma T_S^4$$

$$\frac{P}{4\pi d^2} \times \pi R_E^2 = 4\pi R_E^2 \sigma T_E^4$$

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Linked Comprehension Type **(1 Para × 3Q.) [3 M (-1)]**

(Single Correct Answer Type)

19. Ans. (D)

Sol. Radiation

20. Ans. (B)

Sol. $$U = 4\sigma T^3 \frac{1}{N\left(\frac{2}{\in} - 1\right) + 1}$$ Here N = 50, T = 500 K & $\in = 0.2$

21. Ans. (A)

Sol. $UA\ \Delta T = kA\ \frac{\Delta T}{\Delta \ell}$

$\Rightarrow k = U \times \Delta \ell$

Here $U = 4\sigma T^3 \frac{1}{\frac{1}{\in_1} + \frac{1}{\in_2} - 1}$

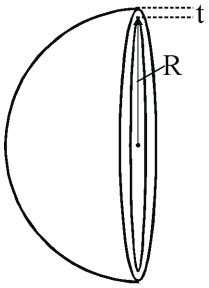
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SECTION-II

Numerical Answer Type Question **4 Q. [3(0)]**
(upto second decimal place)

22. Ans. 300

Sol. $|dP| = \beta\left|\frac{dv}{V}\right|$



Balancing force, $dP\pi R^2 = \sigma 2\pi Rt \Rightarrow \sigma = \frac{dP}{2}\frac{R}{t} = \frac{\beta\gamma\Delta\theta R}{2t}$

where β = bulk modulus
γ = coefficient of thermal expansion
Δθ = change in temperature
R = radius of sphere
t = thickness of sphere

23. **Ans. 450.00**

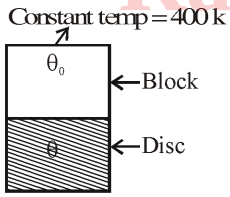
Sol. Temperature as a function of time $T = \frac{1}{10}t°C$

$$\frac{dQ}{dt} = kA\frac{dT}{dx} \Rightarrow \int_0^Q dQ = \int_0^{(10\times60)}\left(\frac{kA}{0.4}\right)0.1t\,dt = \frac{100\times1\times10^{-4}\times0.1t^2}{0.8}\Bigg|_0^{600} = 450$$

24. **Ans. 166.3 sec**

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Sol.



θ = temp of disc
θ_0 = constant temp

Heat input to disc = $\frac{kA(\theta_0 - \theta)}{L}$

Heat utilised by disc = $ms\frac{d\theta}{dt}$

S = Specific heat of disc

$$ms\frac{d\theta}{dt} = \frac{kA(\theta_0 - \theta)}{L}$$

$$\Rightarrow \int_{300}^{350}\frac{d\theta}{(\theta_0 - \theta)} = \frac{kA}{msL}\int_0^t dt$$

$$\Rightarrow t = \frac{msL}{kA}\ell n\left(\frac{\theta_0 - 300}{\theta_0 - 350}\right)$$

$\Rightarrow$ t = 166.32 sec

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25. Ans. 35.00

Sol. Let the equivalent resitance of the wall be R_{eq}

$\therefore$ At steady state, $P=\frac{\Delta T_1}{R_{eq}}$ ...(i), $\Delta T_1=[15-(-10)]^\circ C$

when second heater is switched on $2P=\frac{\Delta T_2}{R_{eq}}$ ...(ii),

$\Delta T_2=[T_{in}-(-15)]^\circ C$ Solving (i), (ii), $T_{in}=35^\circ C$

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) 3 Q. [4 M (0)]

26. Ans. 4

Sol. $mg=\sigma\left(\pi\left(\frac{d_1}{2}\right)^2\right)$

$mg+\frac{mv^2}{\ell}=\sigma\left(\pi\left(\frac{d_2}{2}\right)^2\right)$

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So, $1+\frac{v^2}{\ell g}=\left(\frac{d_2}{d_1}\right)^2 \Rightarrow 1+\frac{36}{12}=\left(\frac{d_2}{2mm}\right)^2 \Rightarrow d_2=4mm$

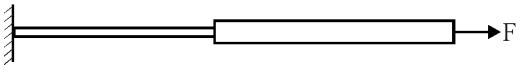
27. Ans. 1

Sol. $V_0(1+\gamma\times 1)-V_0=h(A_0)\Rightarrow h=\frac{V_0\gamma}{A_0}=\frac{1.8\times10^{-4}}{1.8\times10^{-4}}=1\text{ cm}$

28. Ans. 6

Sol. $\frac{F}{A_1}=y\frac{\Delta\ell_1}{2\ell}$

$\frac{F}{A_2}=y\frac{\Delta\ell_2}{\ell}$



$\Delta\ell_1+\Delta\ell_2=\frac{27}{4}$ mm

$\frac{2F\ell}{A_1y}+\frac{F\ell}{4A_1y}=\frac{27}{4}$ mm

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Subjective Type **2 Q. [4 M (0)]**

29.

Sol. (a) As thermal expansion of isotropic solids is similar to true photographic enlargement, expansion of a cavity is same as if it had been a solid body of the same material



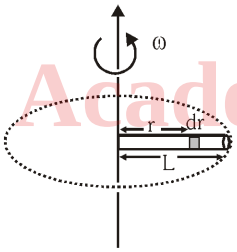
i.e. $\Delta V = V\gamma\,\Delta\theta$

As here V, γ and $\Delta\theta$ are same for both solid and hollow spheres treated (cavity) ; so the expansion of both will be equal.

(b) If same heat is given to the two spheres due to lesser mass, rise in temperature of hollow sphere will be more [as $\Delta\theta = \frac{Q}{mc}$] and hence its expansion will be more [as $\Delta V = V\gamma\Delta\theta$].

30.

Sol. (a) Consider an element of length dr at a distance r from the axis of rotation as shown in figure. The centripetal force acting on this element will be $dT = dmr\omega^2 = (\rho A dr)r\omega^2$. As this force is provided by tension in the rod (due to elasticity), so the tension in the rod at a distance r from the axis of rotation will be due to the centripetal force due to all elements between x = r to x = L



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$$\text{i.e.,} T = \int_r^L \rho A\omega^2 r dr = \frac{1}{2}\rho A\omega^2\left[L^2 - r^2\right] \ldots(i)$$

$$\text{So here } T = \frac{1}{2} \times 10^4 \times \pi \times 10^{-2} \times (400)^2\left[\left(\frac{1}{2}\right)^2 - r^2\right] = 8\pi \times 10^6\left[\frac{1}{4} - r^2\right] N$$

(b) Now if dy is the elongation in the element of length dr at position r then strain

$$\frac{dy}{dr} = \frac{\text{stress}}{Y} = \frac{T}{AY} = \frac{1}{2}\,\frac{\rho\omega^2}{Y}\,[L^2 - r^2]\,dr$$

So the elongation of the whole rod

$$\Delta L = \frac{\rho\omega^2}{2Y}\int_0^L \left(L^2 - r^2\right) dr = \frac{1}{3}\,\frac{\rho\omega^2 L^3}{Y} = \frac{1}{3} \times \frac{10^4 \times (400)^2 (0.5)^3}{2 \times 10^{11}} = \frac{1}{3} \times 10^{-3} m$$

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